INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Apple, OR -- station KDLS, America/Los_Angeles
Committed pair	OSM 706759844 -> 1276373853 Apple / Apple
Facade gap	84.9 m between the two halls
Weather record	42,451 real hourly records from KDLS
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.4502 C at 310 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-05-21 (knife_edge)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 6 of 24 hours with 1 mode change. The reactive incumbent took 9 hours -- 3 more -- but declared 0 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 10 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	6 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	9 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 10 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	12.598	18.0	12.637	1.363
01	mechanical	yes	switch budget	12.345	18.0	12.108	1.333
02	mechanical	yes	switch budget	12.356	18.0	12.052	1.285
03	mechanical	yes	switch budget	12.619	18.0	12.052	1.154
04	mechanical	yes	switch budget	12.034	18.0	11.518	1.400
05	mechanical	yes	switch budget	12.034	18.0	11.518	1.298
06	mechanical	yes	switch budget	12.418	18.0	11.485	1.271
07	mechanical	yes	switch budget	13.695	18.0	12.569	1.401
08	mechanical	yes	switch budget	15.967	18.0	14.822	1.957
09	mechanical	yes	switch budget	17.328	18.0	15.909	2.114

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10  mechanical  no   dry-bulb        19.270    18.0  17.569   2.034
11  mechanical  no   dry-bulb        20.123    18.0  17.019   2.033
12  mechanical  no   dry-bulb        21.741    18.0  19.239   2.125
13  mechanical  no   dry-bulb        22.077    18.0  19.196   2.153
14  mechanical  no   dry-bulb        21.759    18.0  20.349   2.081
15  mechanical  no   dry-bulb        20.859    18.0  18.086   1.774
16  mechanical  no   dry-bulb        19.442    18.0  16.416   1.619
17  mechanical  no   dry-bulb        19.932    18.0  17.628   1.541
18  FREE        yes  -               17.350    18.0  16.485   1.241
19  FREE        yes  -               16.567    18.0  16.485   1.597
20  FREE        yes  -               17.136    18.0  16.518   1.743
21  FREE        yes  -               16.461    18.0  15.935   1.831
22  FREE        yes  -               16.310    18.0  15.359   1.603
23  FREE        yes  -               16.303    18.0  15.359   1.423

bound  = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread
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THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.598 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.345 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.356 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.619 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.034 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.034 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.418 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.695 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.967 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.327 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.270 C against a 18.0 C limit, so it fails by 1.270 C. It would take a limit 1.270 C higher, or a bound 1.270 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.123 C against a 18.0 C limit, so it fails by 2.123 C. It would take a limit 2.123 C higher, or a bound 2.123 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.741 C against a 18.0 C limit, so it fails by 3.741 C. It would take a limit 3.741 C higher, or a bound 3.741 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.077 C against a 18.0 C limit, so it fails by 4.077 C. It would take a limit 4.077 C higher, or a bound 4.077 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.759 C against a 18.0 C limit, so it fails by 3.759 C. It would take a limit 3.759 C higher, or a bound 3.759 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.859 C against a 18.0 C limit, so it fails by 2.859 C. It would take a limit 2.859 C higher, or a bound 2.859 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.441 C against a 18.0 C limit, so it fails by 1.441 C. It would take a limit 1.441 C higher, or a bound 1.441 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.932 C against a 18.0 C limit, so it fails by 1.932 C. It would take a limit 1.932 C higher, or a bound 1.932 C tighter, to change this hour.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.350 C, which is 0.650 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.241 C, and the margin is measured, not chosen: 0.880 C of group-conditional forecast error for this hour of day, plus 0.3606 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.567 C, which is 1.433 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.597 C, and the margin is measured, not chosen: 1.229 C of group-conditional forecast error for this hour of day, plus 0.3677 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.136 C, which is 0.864 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.743 C, and the margin is measured, not chosen: 1.347 C of group-conditional forecast error for this hour of day, plus 0.3959 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.461 C, which is 1.539 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.831 C, and the margin is measured, not chosen: 1.486 C of group-conditional forecast error for this hour of day, plus 0.3450 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.310 C, which is 1.690 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.603 C, and the margin is measured, not chosen: 1.222 C of group-conditional forecast error for this hour of day, plus 0.3807 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.303 C, which is 1.697 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.423 C, and the margin is measured, not chosen: 1.051 C of group-conditional forecast error for this hour of day, plus 0.3728 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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